Chelsea Olyviani Purnomo

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Education

Bina Nusantara University — Jakarta, Indonesia
Bachelor of Computer Science, Data Science Specialization Track (GPA: 3.95/4.00) — *Aug. 2024 – Present*

Research & Publication

Ablation Study on Sentinel-2 and Light Detection Ranging for Mangrove Above-Ground Carbon Estimation
Published, IEEE | view IEEE paper | Github repo — *Sept. 2026*

- Built an XGBoost regression model achieving optimal carbon-estimation accuracy ($R^2 = 0.8779$, RMSE = 6.6687 Mg C/ha)
- Ran a comprehensive ablation study comparing Sentinel-2 multispectral and LiDAR data, showing that fusing LiDAR with GLCM texture features yields the highest model performance

Projects

KosIn | *Python, Streamlit, Scikit-learn* — Jun. 2026
kosin-streamlit | kosin github

- Built an interactive web app to predict boarding-house (kost) rental prices in Jakarta from spatial and property features using a Random Forest model (R2 = 0.8105)
- Selected as one of 22 best teams at Computer Science Festival Kemanggisan 2026

Analysis Indonesia's Renewable Energy | *Python, Seaborn, Matplotlib, Streamlit* — Jun. 2026
streamlit | github

- To evaluates trends in renewable energy growth over time, changes in the national energy mix, the relationship between renewable energy utilization and carbon dioxide (CO_2) emissions, and its connection to economic growth.

FitAI | *Python, Hugging Face* — Dec. 2025
fitAI-huggingface

- Developed FitAI, an AI-powered fitness recommendation platform using Python to generate personalized exercise and meal plans
- Designed a recommendation engine that analyzes comprehensive physical metrics (age, weight, height) as well as user preferences (workout duration, calorie burn, frequency, and experience level)

EIGER Warehouse Capacity & Production Trend Analysis | *Python, Scikit-learn* — Jun. 2025
EIGER-github

- Predict future production volumes beyond historical data (Months M1 to M144) and determine when production would exceed the existing warehouse limit of 25,000 bags/month. To account for complex growth dynamics, non-linear mathematical modeling and numerical root-finding algorithms were implemented.

Experience

Founder & Bootcamp Mentor — Jul. 2025 – Jan. 2026
LogicLab Bootcamp — *Hybrid*

- Founded and led an independent hybrid bootcamp teaching Algorithm and Programming using C programming language to 15+ participants
- Designed and delivered a 13-session curriculum with full supporting materials (slide decks, recordings, exercises, and reference source code)

HIMTI (Informatics Student Association) BINUS — Mar. 2025 – Mar. 2026
Academic Event Committee Member — *Semarang, Indonesia*

- Coordinated 5 main events reaching 500+ attendees, collaborating with 150+ committee members across divisions

Technical Skills

Languages: Python, C, MySQL, PostgreSQL, Java, JavaScript, HTML/CSS
Frameworks: React, Node.js, Vue.js, Streamlit, Prisma ORM
Developer Tools: Git, Github, VS Code, Google Collab, Visual Studio, Eclipse, DevC++, phpMyAdmin, Figma, Cisco Packet Tracer
Libraries: pandas, NumPy, Matplotlib, Scikit-learn, Seaborn

Certifications & Achievements

2nd Place Winner | Lead The Change Business Plan Competition | Link — Oct. 2026

Microsoft Elevate AI Training | Pelatihan Azure AI Fundamentals (AI-900T00-A) | Link — Mar. 2026

Dean's List Recipient 2025 | School of Computer Science | Link — Dec. 2025

Top 10 Finalist | INNOBIZ FST 2025 Business Plan Competition | Link — Nov. 2025

Honorable Mention | The 2025 ICPC Asia Jakarta | Link — Oct. 2025

3rd Place Winner | INBIZ 2025 Business Competition | Link — Sept. 2025